Ex:1
3/07

Setup and Configure a Local Area Network using a Switch and Ethernet cabels in your Lab.

AIM:

To setup and Configure a LAN (Local Area Network) using a switch and 2 Ethernet Cables in your lab.

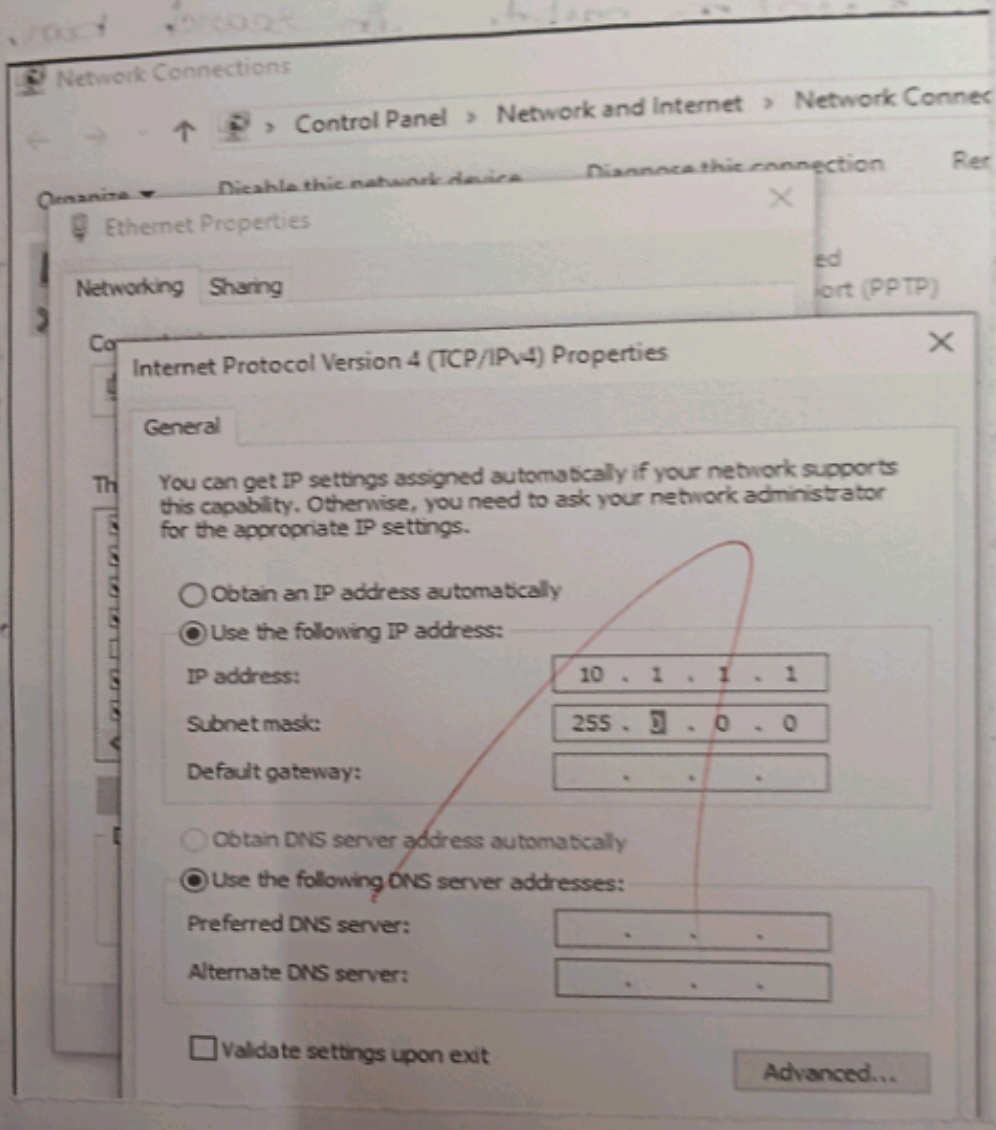
What is LAN:

A LAN is a network that connects devices within a small area, such as a home, school or office. It allows devices like computers, printers and servers to share files, printers and internet access. A LAN switch connects the devices and enables fast and secure communication between them.

How to setup a LAN:

Step1:
Plan and Design an appropriate network topology taking into account network requirements and equipment location.

Step2:
You can take 4 computers, a switch with 8, 16 or 24 ports which is sufficient for networks of these sizes, and 4 Ethernet cables.


Network Connections
Control Panel » Network and Internet » Network Connec
Ethernet Properties
Networking
Sharing
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address:
10 . 1 . 1 . 1
Subnet mask:
255 . 0 . 0 . 0
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...

Step 3:

Connect your computers to network switch via an Ethernet cable, which is a simple a plugging one end of the Ethernet cable into your computer and the other end into your network switch.

Step 1:

Assign Ip address to your PC's

1. Log on to the client computer as Administrator or as owner.
2. Click Network and Internet Connections.
3. Right Click Local Area connection / Element → Go to properties → Select Internet Protocol (TCP/IPV4) → Click on properties → Select use the following ip address option and assign ip address.

Similarly assign IP address to all the PCs connected to switch. PC-IP address: 10.1.1.1 Subnet mask 255.0.0.0 PC-2 IP address — 10.1.1.2, subnet mask 255.0.0.0 PC-3 Ip address 10.1.1.3, Subnet mask 255.0.0.0. PC-4 IP 10.1.1.4 and subnet mask 255.0.0.0.

Step 5:

Configure a network switch

1. Connect your computer to the switch. To access the switch's web interface, you will need to Connect your computer to the switch using a Ethernet cable.

2. Log in to web Interface : Open a web browser and enter IP address of the switch in the address bar. This should bring up the login page for the switch's web interface. Enter the username and password to log in.

3. Configure basic settings : Once you're logged in, you will be able to configure basic settings for the switch.

4. Assign IP address as : 10.1.1.5 , subnet mask 255.0.0.0.

Step6:

Check the connectivity between switch and other machine by using ping command in the command prompt of the device.

Step7:

Select a folder → go to properties → click sharing tab → share it with everyone one the same LAN.

Step8:

Try to access the shared folder from others computers of the network.

90

RESULT:

Thus, the setup and configure a LAN using a switch and Ethernet cables in your lab was created successfully.